Computational evidence for the optimality of Conway–Kochen's 31-vector Kochen–Specker set

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We report a systematic computational search for Kochen–Specker (KS) sets with fewer than 31 vectors in dimension 3, where the Conway–Kochen construction (CK-31) has stood as the smallest known for over three decades and the best proven lower bound is 24 vectors. No sub-31 KS set is found by any of six complementary strategies: MUS-based exact minimization within finite algebraic ray pools, cross-pool mixing, expanded coordinate alphabets, numerical optimization, criticality testing, and triad density analysis. Within each of six algebraic pools, MUS extraction certifies the minimum KS subset size (31 for integers, 33 for Peres/Eisenstein/$\mathbb{Z}[\sqrt{-2}]$, 43 for Heegner-7, 52 for golden ratio). A notable empirical finding is the *modulus-2 boundary*: all tested integer-like alphabets lacking the coordinate ± 2 (such as $\{0, \pm 1, \pm 3\}$) are not KS-uncolorable, consistent with the hypothesis that the cancellation identity $1 + 1 = 2$ is necessary. CK-31 is deletion-minimal: removing any single ray produces a colorable 30-ray set. Vertex merging produces 394 abstract KS-uncolorable 30-vertex graphs; we resolve their realizability within the integer pool via finite constraint satisfaction (all 394 unrealizable within the $\{0, \pm 1, \pm 2\}$ pool).

INTRODUCTION

The Kochen–Specker (KS) theorem [1] demonstrates that quantum measurements in dimension $d \geq 3$ cannot be explained by noncontextual hidden-variable theories. In dimension 3, the theorem requires exhibiting a finite set of rays (directions) such that no consistent $\{0, 1\}$-valued coloring assigns exactly one 1 per orthonormal basis while respecting pairwise orthogonality exclusion. The minimum number of rays needed for such a construction is a fundamental open problem.

Conway and Kochen's 31-vector set [2] uses integer coordinates from the *coordinate alphabet* $\{0, \pm 1, \pm 2\}$—the finite set of values from which ray coordinates are drawn (called "vector components" by Pavicic et al. [6]; the alphabet can be read directly from the coordinate labels on the cube diagrams used to display such constructions [5]). Two rays are orthogonal when $\sum_k u_k v_k = 0$; for coordinates from a finite alphabet, this requires an algebraic relation—a *cancellation identity*—among alphabet elements. For $\{0, \pm 1, \pm 2\}$, the key identity is $1 + 1 = 2$, yielding cancellations such as $1{\cdot}1 + 1{\cdot}1 + (-1){\cdot}2 = 0$. These cancellations create exact orthogonal triples (bases), and when enough bases interlock, no consistent $\{0, 1\}$-coloring exists. CK-31 has stood as the smallest known KS set in dimension 3 for over three decades. The best lower bound is 24 vectors, established independently by Li, Bright, and Ganesh [3] and by Kirchweger, Peitl, and Szeider [8] using SAT-based methods combined with algebraic realizability checking. (Cabello's recent 33-vector Eisenstein construction [9], the "simplest" KS set by the criterion of fewest bases, provides an independent upper bound at 33 for complex coordinates.) Closing the gap between 24 and 31—or proving that 31 is optimal—remains open.

We identify six algebraic number rings known to support KS constructions in dimension 3, each with a distinct cancellation identity producing a distinct ray pool (Table I). Rather than searching over arbitrary ray configurations, we systematically construct alphabets from these algebraic number fields and test every known algebraic source of KS orthogonalities.

METHODS

A *ray pool* is a finite set of projectively distinct rays in $\mathbb{R}^3$ (or $\mathbb{C}^3$), canonicalized so the first nonzero coordinate is positive and coordinates are gcd-reduced. A *triad* is an unordered triple of mutually orthogonal rays (an orthonormal basis up to normalization). The *orthogonality graph* of a pool has rays as vertices and edges between orthogonal pairs; triads are its 3-cliques. Throughout, merging, deletion, and embedding operations act on orthogonality graphs and their triads.

We encode KS-uncolorability as a SAT instance. For a ray set with orthogonal pairs E and triads $\mathcal{T}$, introduce Boolean variables b_v (ray v colored 1). For each triad $\{a, b, c\} \in \mathcal{T}$, add an exactly-one constraint: $(b_a \vee b_b \vee b_c)$ and $(\neg b_\alpha \vee \neg b_\beta)$ for each pair $\alpha \neq \beta \in \{a, b, c\}$. For each orthogonal pair $(u, v) \in E$ not sharing a triad, add $(\neg b_u \vee \neg b_v)$. This extends the standard KS constraint (exactly one per complete basis) by enforcing mutual exclusion on orthogonal pairs that do not complete to any basis in the pool—a strictly stronger condition that is standard in computational KS studies. UNSAT means no valid $\{0, 1\}$-coloring exists.

We tested six complementary strategies, each designed to probe a different region of the search space.

Strategy 1: SAT-verified minimization. For each of the six algebraic ray pools (Table I), we applied three heuristic methods: (a) clause-level MUS extraction,

(b) greedy deletion with 2000 random trials plus deterministic orderings, (c) swap-based local search. The SAT encoding uses Glucose4 [4] via PySAT.

Strategy 2: Cross-pool mixing. We merged rays from different algebraic pools, deduplicated by canonical form, and tested whether cross-pool orthogonalities enable smaller KS sets. All 15 pairwise combinations and the full 477-ray union of all six pools were tested with 500 greedy trials each.

Strategy 3: Larger alphabets. We tested 30 expanded alphabets: integer $\{0, \pm 1, \ldots, \pm k\}$ for $k = 3, 4, 5$; extended Peres, Eisenstein, $\mathbb{Z}[\sqrt{-2}]$, and Heegner-7 pools; mixed-field alphabets; completion-expanded pools; and Pisot-number fields (plastic, tribonacci, silver ratios).

Strategy 4: Numerical optimization. Four non-algebraic methods: simulated annealing ($n = 28$–30 rays, 2×10^5 steps), CK-31 perturbation (random ray replacement, 1000 attempts), orthogonal completion from random seeds, and soft-tolerance annealing.

Strategy 5: CK-31 criticality. We tested whether CK-31 can be reduced by removing k vectors for $k = 1, \ldots, 12$, exhaustively checking all $\binom{31}{k}$ subsets for $k \le 8$ and sampling 10^6 random subsets for $k = 9, \ldots, 12$.

Strategy 6: Triad density analysis. For each pool, we sampled random subsets of varying size and determined the threshold cardinality at which KS-uncolorable subsets first appear. This characterizes the "rarity" of KS sets within their own pools.

TABLE I. Minimum KS subset size per pool (Strategy 1). "SAT" is the minimum found by MUS extraction (all MUS trials returned this size). "Cross" is the minimum found when that pool is included in any pairwise or full cross-pool merge (Strategy 2).

Pool	Rays	Triads	SAT	Cross	Larger
Integer	49	26	**31**	31	31
Eisenstein	57	22	33	33	33
Peres	49	16	33	33	33
$\mathbb{Z}[\sqrt{-2}]$	49	16	33	33	33
Heegner-7	145	42	43	43	43
Golden ratio	205	166	52	52	52

TABLE II. Selected expanded alphabets (Strategy 3). "KS" indicates whether the pool is KS-uncolorable; "Min" is the best greedy result in 300–500 trials.

Alphabet	Rays	Min	Note
$\{0, \pm 1, \pm 2\}$	49	31	Baseline (CK-31)
$\{0, \pm 1, \pm 2, \pm 3\}$	145	31	Absorbs to CK-31
$\{0, \pm 1, \pm 2, \pm 3, \pm 4\}$	289	36	Heuristic limit
$\{0, \pm 1, \pm 3\}$	49	—	Not KS
$\{0, \pm 1, \pm 4\}$	49	—	Not KS
Eisenstein c=3, n≤6	345	31	Absorbs to CK-31
Peres $+ \pm 2$	109	31	Absorbs to CK-31
Heegner-7 $+ \pm 2$	253	31	Absorbs to CK-31
Golden $+ \pm 2$	637	33	No absorption
$\{0, \pm 1, \pm 2, \pm i\}$	127	31	Mixed-field

RESULTS

MUS-based minimization (Strategy 1). The KS coloring formula for each full ray pool is unsatisfiable (the pool is KS-uncolorable). A Minimal Unsatisfiable Subset (MUS) of this formula identifies a minimal KS-uncolorable ray subset. For each pool, we performed 1,000–5,000 independent MUS extractions (randomized deletion with PySAT/Glucose4); all returned subsets at or above the reported minimum size. For the integer pool, all 5,000 extractions returned size 31; for Eisenstein, Peres, and $\mathbb{Z}[\sqrt{-2}]$, all returned size 33. As additional confirmation, exhaustive neighborhood search around each minimal set (all single-ray removals, all ray swaps, all two-ray removals) found no smaller KS set. From the integer pool, we found six distinct minimal 31-sets, collectively using 37 of 49 rays. These six sets exhibit sharp norm stratification: 13 rays with $\|v\|^2 \le 3$ (the $\{0, \pm 1\}$ sub-alphabet) form a core present in all six sets; 12 rays with $\|v\|^2 = 9$ appear in none. No two 31-sets are connected by a single-ray swap (minimum swap distance: 5 ray replacements), yet all share 25–26 rays (Jaccard ≈ 0.71). (We cannot rule out the existence of additional minimal 31-sets not reached by our randomized MUS extraction.)

Cross-pool mixing (Strategy 2). Despite significant cross-pool orthogonalities (e.g., Integer + Golden shares 78 cross-triads), every minimized set draws rays from a single pool only: any combination including Integer yields 31 (all Integer); without Integer, 33. The 477-ray union minimizes to 31 Integer rays. Note that greedy deletion is biased toward large basins of attraction, so mixed-pool solutions cannot be ruled out.

Larger alphabets (Strategy 3). We tested 30 expanded alphabets (Table II). Every extension that includes ± 2 converges to 31 using integer rays. Greedy minimization is heuristic, so for large pools (e.g., $\{0, \pm 1, \ldots, \pm 4\}$ with 289 rays reaching only 36 in 500 trials) the true minimum may be lower; however, every pool containing the $\{0, \pm 1, \pm 2\}$ subset found 31 when given sufficient trials.

The modulus-2 boundary. The alphabets $\{0, \pm 1, \pm 3\}$ and $\{0, \pm 1, \pm 4\}$—which lack ± 2—are *not KS-uncolorable at all*, despite generating 49 rays with 10 triads each. This is consistent with the hypothesis that the cancellation identity $1 + 1 = 2$ is necessary for KS-uncolorability from integer-like alphabets. More broadly, across all tested fields, KS-uncolorability is observed only when one of two cancellation mechanisms is present: *modulus-2 cancellation* (the generator x satisfies $|x|^2 = 2$, as in $|\sqrt{2}|^2 = 2$, $|\sqrt{-2}|^2 = 2$, $|\alpha|^2 = 2$; the integer case $1+1=2$ is the degenerate additive instance) or *phase cancellation* (a root-of-unity sum $1+\omega+\omega^2=0$). In all tested rings where neither mechanism is available—including those with $|x|^2 \geq 3$—the pool is colorable despite containing orthogonal triples. Three higher-degree fields (plastic ratio, tribonacci, silver ratio) also produce only colorable pools, extending this observation beyond quadratic fields. The boundary is empirical: cancellations not involving 2 may exist in untested fields.

Numerical optimization (Strategy 4). As a negative control, we tested four non-algebraic methods: simulated annealing, CK-31 perturbation, orthogonal completion, and soft-tolerance annealing. All fail, producing at most 5–6 orthogonal pairs and zero triads. This is expected: exact orthogonality is measure-zero in continuous space, so continuous optimization cannot find KS sets without algebraic structure. The result confirms that algebraic methods are necessary, not merely convenient, and is not independent evidence against sub-31 KS sets.

Vertex merging and realizability. We generate 30-vertex candidate KS graphs by merging non-orthogonal ray pairs in CK-31 (identifying two vertices into one that inherits both neighborhoods). Merge preservation of KS-uncolorability follows from monotonicity: if the merged graph were colorable, assigning the merged vertex's color back to both original vertices would yield a valid coloring of the original, contradicting its uncolorability. All 394 non-orthogonal pair merges of CK-31 thus produce abstract 30-vertex KS graphs. The key question is realizability: can any of these abstract graphs be embedded as rays in $\mathbb{R}^3$ or $\mathbb{C}^3$? Z3 [7] returns "unknown" on all 394 $\mathbb{R}^3$ realizability queries. Reformulating as a finite SAT encoding (Boolean assignment of vertices to the 49 pool rays, with orthogonality and injectivity constraints) resolves all 394 instances in under 1 s each: all are *unrealizable* within the integer pool. This illustrates the realizability bottleneck: combinatorially viable 30-vertex KS graphs exist, but none embed in the $\{0, \pm1, \pm2\}$ pool. We have not tested realizability of these 394 graphs in other pools (Eisenstein, Heegner-7, etc.) or in $\mathbb{C}^3$; such embeddings remain open.

CK-31 deletion-minimality (Strategy 5). CK-31 is deletion-minimal: removing any single ray produces a colorable 30-ray set (all 31 removals tested exhaustively). Since KS-colorability is hereditary under deletion (any subset of a colorable set is colorable), deletion-minimality implies that *every* proper subset of CK-31 is colorable. As a computational diagnostic, we also tested larger deletions (Table III): at $k=7$, where 24 rays remain—the Li–Bright–Ganesh lower bound—all 2,629,575 resulting 24-ray subsets are colorable. This confirms that no 24-ray sub-configuration of CK-31 is a KS set, though this already follows from deletion-minimality.

Triad density threshold (Strategy 6). In 500 uniform random samples per size (an exploratory diagnostic, not a precise threshold estimate), no KS-uncolorable subset of the integer pool is observed below $n=42$ out of 49 rays (upper bound on prevalence: $p \lesssim 0.006$ at 95%); for the Eisenstein pool, $n=48$ out of 57; for Heegner-7, none at any sampled size up to $n=75$ out of 145. Among all tested constructions, CK-31's minimum achieves the highest pair density (15.3% of $\binom{31}{2}$ pairs are orthogonal). These results show that triad count alone does not determine uncolorability—the topological interlocking of constraints matters.

DISCUSSION

Our results provide computational evidence bearing on four aspects of KS minimality in dimension 3.

First, *within each of six finite algebraic pools, MUS extraction consistently finds a sharp minimum* (Table I). Combined with the rigidity of CK-31 [10] (uniqueness up to unitary equivalence), this provides strong evidence that 31 is optimal within integer coordinates and 33 within the Peres, Eisenstein, and $\mathbb{Z}[\sqrt{-2}]$ pools. The integer pool contains at least six distinct minimal 31-sets (found across 5,000 MUS extractions), all sharing a 13-ray invariant core and excluding all highest-norm rays.

Second, *the tested algebraic pools appear self-contained*: in all greedy cross-pool trials, no mixing produced a smaller KS set than exists within a single pool. (This is heuristic; exhaustive cross-pool search is infeasible.)

Third, *CK-31 is deletion-minimal*: every proper subset is colorable (by monotonicity from single-ray deletion).

Fourth, *density arguments appear insufficient for a proof of optimality.* KS sets are extraordinarily rare even within pools that contain them. The uncolorability depends on the precise topological interlocking of basis constraints—a property that cannot be captured by counting triads or pairs alone. This suggests that closing the 24–31 gap will require realizability-based arguments (extending the Li–Bright–Ganesh framework) rather than combinatorial density bounds. We note that the gap might also be narrowed from below: the current lower bound of 24 is not known to be tight, and improving it to, say, 28 or 29 via stronger SAT encodings would

TABLE III. Extended criticality of CK-31. For $k \leq 8$, all $\binom{31}{k}$ subsets tested exhaustively; for $k=9$–12, 10^6 random subsets sampled. All subsets colorable.

k	Remain	Subsets	Method
1–4	30–27	36,456	Exhaustive
5	26	169,911	Exhaustive
6	25	736,281	Exhaustive
7	24	2,629,575	Exhaustive
8	23	7,888,725	Exhaustive
9–12	22–19	4×10^6	Sampled

be equally significant.

Working in $\mathbb{C}^3$ does not help: our complex pools all minimize to ≥ 33.

We do not prove that 31 is optimal. Among all tested two-element alphabets $\{0, \pm 1, \pm x\}$, the cancellation identities producing KS-uncolorable pools are $x = 2$, $|x|^2 = 2$, $|x|^2 = x + 1$, and $x^2 + x + 1 = 0$—none achieving fewer than 31. A density trend is observed: simpler identities produce denser orthogonality structures (integer: 0.53 triads/ray, minimum 31; golden ratio: 0.20, minimum 52), suggesting that sub-31 density from a more complex identity would be difficult. The Cortez–Morales–Reyes invariant $N(S) = \mathrm{lcm}\{\|v\|^2\}$ [11] provides an independent perspective: CK-31 has $N = 30$, while the Eisenstein island achieves $N = 6$—exactly the minimal ring $\mathbb{Z}[1/6]$ of their theorem.

Our finite pool SAT encoding is directly applicable to the realizability-based lower-bound programs of Li, Bright, and Ganesh [3] and Kirchweger et al. [8], both bottlenecked by Z3 returning "unknown" on embeddability queries above order 20. By reformulating realizability within a finite ray pool as pure Boolean satisfiability, the encoding resolves instances in under 1 s that defeat Z3, and could serve as a "quick accept" screen in the LBG pipeline.

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